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As preprints grow in popularity, one researcher takes stock of the benefits and challenges for the AI ethics field.

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Preprinting in AI Ethics: Toward a Set of Community Guidelines

THE FAST-MOVING, DYNAMIC world of artificial intelligence (AI) stands in stark contrast to the slow-moving, conservative world of academia.¹¹ This is particularly clear in the world of AI ethics, where in addition to the industry-academia contrast we also have the meeting of very different academic disciplines, including computer science, philosophy,

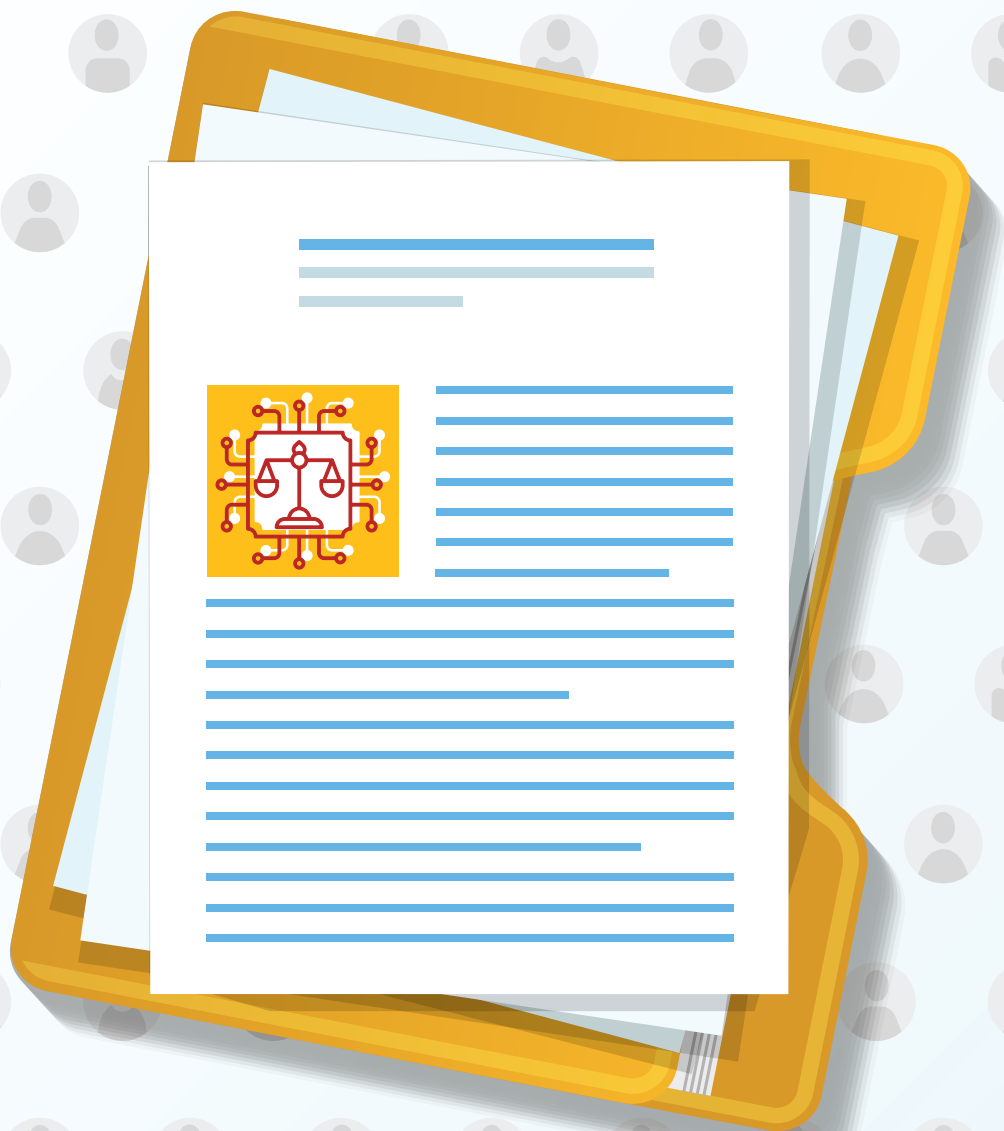
ethics, and social sciences. The traditions, norms, and values of these disciplines are often at odds with one another, making interdisciplinarity challenging. Take, for example, *preprinting*, the practice of quickly disseminating research before potentially—but not necessarily—seeking publication in traditional academic journals.^a Interdisciplinary conflicts appear when, for example, researchers from a computer science background, where rapid publication of preprints on servers such as arXiv is the norm,² meet researchers from the social sciences and humanities, where this is less common.^{1,30}

AI ethics emerged as an active research field over the past decade, but it has a long history and a strong overlap with other fields of applied ethics, such as computer ethics.²⁵ AI as a technology is not new either,²³ nor are most of the issues with which it grapples, such as privacy, bias, and programmers' responsibilities. The general scientific community has been portrayed as a "republic of science" by Michael Polanyi;²¹ some, though, have noted that this republic consists of partly overlapping domains—"fiefdoms of

^a The definition of a preprint is contentious,³⁴ but suffice it to say a preprint is an article published informally, openly, and freely, yet in accordance with scholarly or scientific standards and with the ambition of contributing to the research community.¹⁶

» key insights

- **AI ethics lives at the intersection of diverse disciplines, making rapid knowledge-sharing both vital and challenging.**
- **Preprints can foster speed of dissemination, open critique, and novel ideas, but also risk amplifying misinformation, eroding community standards, and misdirecting policy.**
- **Different academic and industry norms within AI ethics underscore the potential benefit of shared preprinting guidelines and ethical safeguards, which could help foster mutual understanding and trust and contribute to better AI solutions and policies.**



expertise.”¹⁰ The norms of disciplines vary greatly,^{1,30} and a struggle to identify the community norms around preprinting in AI ethics is currently developing, as different cultures clash in this border fiefdom.

In this article I focus on the practice of preprinting, including the role and status of preprinting compared with traditional journal publication and other forms of dissemination, such as blogging and mainstream media contributions. Preprinting has many advantages, such as rapid dissemination of and feedback on ideas, openness, the bypassing of quarrelsome gatekeepers in the traditional publishing system, and the opportunity to publish novel ideas in new formats.^{4,7,27,35} In particular, it is often hailed as a beneficial practice for early-career researchers in need of quickly establishing a track record.^{2,19}

But before we accept the idea that preprinting has many benefits and should be adopted by all, certain challenges should be noted. These include the dangers of bad science, misdirection of policy and practice, disinformation, and information overload.^{12,14,15,36} One important issue concerns how the growing prominence and legitimacy of preprints could more easily allow actors to use preprints in strategic efforts to influence and shape various ideals, norms, or systems. This could, for example, happen through the spread of fake science and manufactured scientific support for conspiracy theories. In addition, the seeming “democratization” of authority through preprinting can be deceptive,¹² as I will argue that departing from traditional norms of scientific publishing with peer review might in fact be most beneficial for those who already wield power.

I begin with a presentation of the benefits and challenges of preprinting based on a review of the literature, including some thoughts on how these apply to AI ethics specifically. Then I suggest some broad best practices around preprinting with the aim of sparking discussion within the community.

Why Preprint?

There are many good reasons to preprint, and a quick review of the literature on preprinting in different



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disciplines shows clear similarities. In the following, I highlight some of the main *benefits* highlighted in the literature, followed by a brief discussion of how AI ethics researchers specifically might benefit from preprinting. The benefits are listed according to how often and how strongly they are reflected in the literature, with the most strongly emphasized discussed first. Here, I largely exclude the potential negative effects and costs related to preprinting, as these will be discussed in the next section.

Speed and effective dissemination and feedback. A primary benefit of preprinting is that it allows for the rapid and effective dissemination of new ideas.^{4,7,9,12,27,30,36} Many blame the publishing system and the time-consuming process of peer review for the need to preprint to get one’s ideas out there more quickly.^{9,12,27,32} The challenges of lengthy peer review and low acceptance rates are more severe in the humanities than in other disciplines, some argue.³² Although peer review certainly has benefits, the peer-review system has been thoroughly criticized in the broader literature, where it is seen as essentially flawed, inconsistent, slow, and ineffective.¹³ Preprinting is seen by some as a partial solution, or at least a way to bypass this system. Rapid dissemination can also help researchers “maintain enthusiasm” for their work,¹⁵ as the traditional review process is so slow that it can make the work feel distant when revision and potential publication occur *long* after the ideas were generated and written about.

Scientific deliberation and wide review. Another major benefit pertains to how preprints can promote broad scientific deliberation, in which entire communities engage in reviews and examinations of early-stage research.^{2,4,7,9,14,27,36} Peer review often entails vetting by a very limited number of scholars, while preprints could, in theory, allow for vetting by a much larger audience.² Peer review is often assumed to improve the quality of research and help identify good science,^{8,13} so one might expect that broader review processes could help improve science and foster researcher learning.²⁷ The broad and rapid review that potentially ensues from preprints can also help quickly evaluate contro-

versal results.⁴ For example, in 2023, a group of researchers released a preprint presenting a room-temperature superconductor¹⁷—something that, if real, would have been a huge leap in science—generating massive activity and a number of replication efforts. This was soon followed by other preprints that seemed to have debunked the original paper.²⁰ This all took place within the span of a couple of weeks. While these sorts of events might be assumed to promote an interest in science and generate awareness, it could also be argued that this preprint, and the hype surrounding it, was wasteful and unfortunate, and that a proper vetting by peers before publication would have prevented the whole situation.

The review process for preprints varies, and includes, for example, social media discussions, counter-preprints, or straightforward peer-review reports. The latter can be found in a range of services aimed primarily at the life sciences, such as eLife, Peer Community In, and Review Commons, but the mechanics of peer review could be developed equally well for AI-ethics-relevant servers. For example, PREReview, a preprint review service, also provides integration with arXiv.

Though broad review is a potential benefit of preprints, it must be noted that just posting a preprint comes with no guarantee that it *will* be widely reviewed or reacted to. Though some preprints get a lot of attention, the vast majority are likely not vetted in any meaningful sense.

Author empowerment and early-career researchers. Various stakeholders, such as researchers, funders, journals, and universities, can have different interests in and evaluations of research.² Researchers are often touted as major beneficiaries of preprinting, with the literature emphasizing various forms of author empowerment.³³ One benefit is that preprinting allows for self-archiving and both the documentation and dissemination of one's work.¹ It also potentially helps authors retain the rights to their research and control its use.¹⁹

While researchers in general could see personal benefits from preprinting, the literature stresses the benefits

for early-career researchers.^{2,7,9,19,27} Preprints can be used in resumes and job and grant applications, and can help papers get early citations.^{9,19} Furthermore, they help with visibility, networking,²⁷ and catalyzing collaboration.¹⁵ In short, preprinting increases exposure to both scholarly and popular audiences.²⁹

Bypassing gatekeepers and the spread of novel work. Science could be argued to symbolize and promote liberal-democratic ideals, but it simultaneously challenges the same “through its exclusivity and elitism.”³ Though there are many different academic journals and outlets in which to publish, some argue that formal publication and peer review serve a problematic gatekeeping purpose that disqualifies novel work and ideas, null results, and certain formats and styles of research. They can also be a significant barrier to groundbreaking interdisciplinary work that will suffer when evaluated by specialists in established journals.¹⁰ Preprinting can therefore help researchers bypass gatekeepers and publish novel ideas, in both traditional and novel formats.^{4,14,35} Vuong,³⁵ for example, offers several reasons to preprint, based on his own experience. One was that he and his coauthors disagreed with the reviewers and editors, deciding to preprint rather than compromise by revising the paper according to reviewer recommendations. Though gatekeeping and revision after peer review might improve the quality of research, they can also remove some of the original thinking found in preprints.³⁵

Open science and promoting access to knowledge. The notion of open science is increasingly prevalent in the research community. Preprinting can contribute to this by promoting access to, and potentially improving the quality of, science.^{19,27,30} Assuming that increased access allows more actual and potential researchers to join in scientific endeavors, one might expect an increase in quality. One might also, however, imagine a situation in which openness changes and possibly undermines community standards and quality assurance procedures, which could be detrimental to the quality of science.

Related to the costs and benefits of bypassing gatekeepers, preprinting

can also help “democratize” science and scientific authority.¹² Openness and free publication allow researchers to bypass the structural injustices prevalent in the publishing system, where the ability and willingness to pay to publish, for example, prevail. This system gives researchers from well-off institutions significant visibility benefits. It also entails that only one aspect of the open access ideal—access to published research—is achieved, while access to open *publication* is reserved for the privileged. Preprinting potentially eliminates this challenge and places everyone on more equal footing.

Documenting precedence, priority, and process. Finally, preprinting provides a public and transparent foundation of precedence and priority in research by documenting the discovery and research processes.^{2,4,7} It provides a transparent record of a researcher's work,⁴ and “a fair and straightforward way to establish precedence.”⁷ It also helps document the research *process* from early to final-stage research, and provides insight into and proof of the co-creation process that occurs when reviewers read and comment on the various revisions preceding the publication of research.^{2,16}

The benefits as they pertain to AI ethics. All of the benefits of preprinting apply to the AI ethics discipline, but it is likely that *technically* focused researchers derive greater benefits than the more philosophically oriented members of the field.³² The need for speed—for both author and audience—is arguably greater when research is about new methods, algorithms, and so on than when it focuses on, for example, new ways of understanding the differences between and applications of ethical frameworks such as consequentialism and deontology. But social science and humanities scholars also provide vital analyses of the implications of new technologies—that is, the need for speed might be just as important for a philosopher as it is for a computer scientist.

For individual researchers, however, the benefits related to rapid dissemination, greater visibility, and a track record for applications, citations, and the like will apply regardless of whether the *audience* needs

their research. This could lead to a situation where philosophically oriented researchers working in the same field as more technically oriented researchers start adopting the same preprint practices to stay competitive—they are, after all, competing in roughly the same job market. This is not to disregard the fact that the top journal publications are still valued in many instances, meaning that the issue is more about *balance* than it is an either-or question.

But if we agree that a joint or at least similar approach by all working in the same field would be beneficial, whose norms and practices should be adopted? One group might unilaterally adopt the other group's norms. Or we could seek a compromise where both sides adjust, adopting a practice that makes the best use of preprinting while seeking traditional journals or other outlets for work best suited to those venues.

A major benefit for AI ethics is how preprinting allows researchers to overcome the difficulties that come with working in a new and interdisciplinary field. There are few specialized, high-ranking outlets where AI ethics research gets peer reviewed by traditional specialists, so this provides a good reason to preprint. It also underscores the need for new journals and outlets. This is particularly important in the field of AI, as the technology has been unleashed on societies, leaving many stakeholders—including policymakers—scrambling to find good responses. AI ethics could inform their efforts, but this would be more effective if the community had common practices for disseminating its work.

The Challenges of Preprinting in AI Ethics

Preprinting provides a wide range of potential benefits, but what does the flip side of the preprinting coin look like? Here, I summarize some of the challenges emphasized in the literature, while adding a couple of potential objections not particularly well covered in extant research.

Junk science let loose. An obvious challenge presented by a lack of peer review is that it allows for the dissemination of poor-quality research.^{14,15} While erroneous and misleading re-

sults can be published quickly, they will also stay published, despite subsequent reviews and refutations. A recent example is the preprint suggesting that ChatGPT suddenly performed more poorly—and at times *significantly* more poorly—over time.⁶ The paper received a lot of attention at the time of publication, including coverage in major news outlets, despite other researchers pointing out major problems with its claims, such as its failure to note that code performance *increased* if one removed new surrounding code.^b

That preprints can be influential is quite obvious. For example, a paper by researchers from Microsoft—an investor in OpenAI—found “sparks” of artificial general intelligence in OpenAI's GPT-4.⁵ At the time of this writing (January 2024), the paper has 1,183 citations—quite good for a preprint from April 2023.

Journals, however, are subject to the same challenges. Peer review is practiced very differently across fields and publications. And there is a proliferation of new journals—including predatory ones—that offer no or only superficial peer review, yet portray published research as if it has been through rigorous peer review.

Misdirection of policy and practice. Early results might be instrumental in responding rapidly to new situations, such as COVID-19.³⁶ However, we have also seen that preprints can be of poor quality and misleading. When such research gets the wrong people's attention, it could lead to misdirection of policy and practice^{12,15,36} and a general “confusion and distortion” of the processes that lead to proper, evidence-based policy responses to social problems.²⁸ In the case of COVID-19, for example, a plethora of preprints were released about the methods and likelihood of transmission, and how to treat and prevent the spread of the virus.³⁶ One suggested that the drug Ivermectin might be effective in combatting its effects, leading to the drug being administered despite a lack of evidence, with subsequent findings showing it was harmful and not effective.³⁶

b https://twitter.com/Si_Boehm/status/1681801371656536068?s=20

Preprints could lead some to, for example, base policies on early or uncertain results; however, it will at times be a question of *some* uncertain knowledge vs. *no* knowledge. In this case, preprints, when evaluated and used with caution, could play an important role.

Information overload. While there are quite a few journals out there—more than most of us are able to keep track of—preprints bypassing journals altogether could exacerbate information overload.^{12,14} Researchers might struggle to find the most relevant research, as might the general public. More importantly, and as Sheldon²⁸ argues, the primary curator of important science today is the *media*, which mainly depends on established outlets and their publication and dissemination procedures to fulfill this function. Moving to preprints, he argues, risks highlighting bad science and neglecting other important science. Though this concerns media practices, the research community must actively work with and educate the media on what the good channels are, and how to distinguish solid research from that of more uncertain quality. Though the proliferation of journals could create certain challenges for the traditional publication model, some journals will have strong, long-established reputations that they can use as a stamp of approval. To avoid complete information overload, it might be important to communicate these reputations to stakeholders beyond academia.

Some argue³⁰ that researchers and evaluators of science are already faced with more research than they can digest. As a result, there is fear that increased use of preprints will become a time sink for researchers and grant reviewers—due to both the volume of research *and* the extra need for caution and careful evaluation of preprints.²

Disinformation and conspiracy theories. Distinguishing pseudoscience from *science* has long been a concern in the scientific community, with some seeing the spread of pseudoscience through non-academic channels as a source of confusion for the general public.¹¹ As such, preprints might also be seen as a new and particularly

effective source of confusion. Journals arguably serve a social function by having editors and reviewers who vet research. This model is problematic and under pressure, yes, but it is still far better than nothing. When this control is bypassed, we are more vulnerable to disinformation in the form of science-like research.^{12,14}

This could take place accidentally through the spread of junk science, but it could also occur through strategic efforts aimed at promoting or undermining various ideals, norms, or systems. While this has not received attention in the literature, science could be subject to the same type of dynamics found in social media, where “troll farms,” for example, are mobilized for political purposes.³⁷

There is also the potential for the dissemination of articles that support or give rise to conspiracy theories,^{12,36} which did, in fact, occur during the COVID-19 pandemic, when a preprint arguing for the “uncanny” similarities between COVID-19 and human immunodeficiency virus spurred conspiracy theories.³⁶ These challenges will become increasingly pressing as those wanting to promote disinformation and conspiracy theories make full use of generative AI.²⁶ Generative AI can easily be used to construct misleading or outright fake articles—even with fake data—that will likely pass most preprint servers’ relatively low bar of quality control. (Though it could also pass the bar of certain journals, I’ll argue that the probability of such articles being identified and rejected will generally be somewhat higher.)

Scooping and problems with subsequent publication. An often-mentioned challenge in the literature is the risk of being *scooped*.^{4,14,15,27} However, most papers mention this researcher fear as being largely unfounded, arguing that preprints do *not* make one susceptible to scooping due to the paper trail and public nature of documented results.⁴ A related concern is the fear of losing out on publication opportunities if one preprints.^{1,9,12,14,27} This challenge, though, has been mitigated by drastic changes in publisher policies, as most now allow some form of preprinting.^{4,7}

Challenges related to power and



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authority. Open science leads to more publications, and preprinting is, as we have seen, perceived as a good strategy for researchers.¹⁹ One objection to this perspective, however, is that the radical equality in the preprint world makes the playing field *less* even and *less* welcoming for unknown and early-career researchers. The idea behind blind peer review is that name and face should not matter—that, for example, Alan Turing and a rookie researcher would be equally positioned in the struggle to publish in a top journal and benefit from the “heuristic cues of a journal’s reputation, selection, and peer-review processes.”³⁰ Much research shows that anonymous peer review is far from perfect, of course,¹³ but might still be better than a situation in which one’s name and reputation are immediately obvious and just about the *only* cue readers have to go by. Preprinting, one might argue, favors the strong—*not* the weak.

There are also different sources and domains of power in AI ethics. While the traditional and institutional university sector could be seen as providing one source of authority, in the world of AI ethics, a social media presence and recognition by peers are arguably just as important. Those marginalized from traditional institutions might perceive themselves as outsiders, but with a large number of social media followers, they inevitably wield power and must be considered to be powerful actors in the fiefdom of AI ethics. Social media provides a new form of power in the research community, and X/Twitter mentions, for example, are linked to the number of citations and arXiv downloads.^{29,36}

Another way of using preprints to further power and influence is through unblinding journal reviews. If someone like Luciano Floridi, for example, submits a previously preprinted article for review in a publication, any theoretical notion of the journal’s blind review standards will evaporate. One might argue that reviewers and editors are sufficiently professional to disregard name recognition; however, this position seems naive. Grove provides a relevant anecdote about the workings of power in the academic system:

*When Lord Rayleigh, already a highly reputed scientist, submitted a paper to the British Association for the Advancement of Science in 1886, his name was somehow omitted and the paper was rejected. When its authorship was discovered, it was at once accepted.*¹¹

Preprinting unveils power, and one might argue that blind reviews provide an imperfect, yet not wholly ineffective, veil that primarily helps those without strongly established reputations. Earlier, preprinting was mentioned as a force of democratization;¹² these considerations indicate that it could also have elitist and undemocratic effects.

Research ethics. Preprinting has the additional downside of potentially allowing research without due ethics approval, conflict of interest statements, and so on—routinely required by publishers and journals—to be disseminated.⁹ Bypassing such routines might, of course, be a problem because it leads to bad science, but it could also promote a situation in which researchers take greater risks, and where individuals become more exposed to both unethical research practices *and* potential privacy breaches following preprinting. This will largely not be the case when researchers publish preprints with an eye toward subsequent journal publication, as this will require properly dealing with ethics committees and approval in the early stages of the research, as well as in the final journal article. However, if, over time, preprints become legiti-

mized and more accepted, it could allow those not mandated by their institutions to get comprehensive ethical clearance to both conduct and publish such research. In the traditional system, industry actors not bound by the same rules and guidance as, for example, university researchers, would be incentivized to follow traditional research ethics standards to be able to publish in top journals.

Manipulation of metrics and sector incentives. Finally, a minor objection concerns how preprints can be used to boost a researcher's publication count and citation score. For example, preprints are indexed by Google Scholar, thus opening the possibility of inflating both article and citation counts through prolific preprinting. Preprints can therefore be used to game and exploit the various indicators increasingly prevalent in the world of research.

The challenges as they pertain to AI ethics. The negative aspects of preprinting all apply to AI ethicists, but to varying degrees, depending on the topic and type of research.

First, there is a lot of hype around AI; deployment of AI systems is moving very rapidly, and policymakers and other stakeholders are scrambling to identify the best responses. This suggests that the speed and flexibility of preprinting could be particularly valuable in this field, but it also shows that the potential for harm is significant. Low-quality research will always be detrimental to the scientific community, but it is arguably more harmful

when it is in danger of being misinterpreted *or* strategically misused to inform policy or practice in harmful ways. In addition, low-quality research contributes to the crowding of the marketplace of ideas and risks increasing information overload and burying high-quality research.

Second, the practice of acknowledging and citing preprints could promote the acceptance of unethical research conducted without sufficient safeguards for either society or individuals. It is therefore incumbent upon the community as a whole to be restrictive when citing and using research that does not adhere to the ethical standards one strives for, and this requires careful attention to data availability, protocols, statements, and so on.

Third, the community should be aware of the potential for powerful actors and groups to bypass community institutions and change the norms of the fiefdom by first gaining status and then simply preprinting, without an eye to publication. This would undermine whatever quality control we have in the existing faulty system, and could also lead to a situation where preprinting becomes standard practice, as others start citing and using preprints actively in their own research.

Finally, as preprints can be abused for disinformation and other manipulation efforts, it is important that the community stay alert to such efforts to influence either the scientific community *or* the public through dissemination of propaganda masked as scientific preprints.

Toward Best-Practice Preprinting in AI Ethics

The preceding sections have detailed a range of benefits and challenges related to preprinting, summarized in the table.

At a time when disinformation,³⁷ polarization,³¹ and a lack of trust in the scientific community in large groups¹⁸ are prevalent, it is important that the scientific community considers its role and responsibilities and actively engages in the shaping of our academic culture and practices. This is no less relevant for the field of AI ethics, where we deal with technologies that are rapidly implemented, with major implications for individuals, businesses, and

Table. Summary of the benefits and challenges of preprinting.

Aspect	Benefits of Preprinting	Challenges of Preprinting
Speed and Efficiency	• Rapid dissemination of findings • Early feedback and engagement from the community	• Risk of spreading preliminary results that may be flawed • Potential for rapid dissemination of bad science
Accessibility	• Enhances visibility of research, especially for early-career researchers • Bypasses traditional gatekeepers	• Can lead to information overload • Difficulties in navigating the increased volume of literature
Review Process	• Broader scientific deliberation and review • Engages a wider audience in the critique process	• Lack of formal peer review can undermine quality control • High variability in reviewer expertise
Empowerment	• Authors maintain control over their work • Helps establish track records and boost citations	• May disproportionately benefit established researchers with wider recognition and networks
Openness	• Promotes open science and transparency • Facilitates collaboration and sharing of ideas	• Challenges in maintaining quality standards • Potential misuse of preprints for promoting pseudoscience

governments. As mentioned, there is much hype concerning AI, and a high demand for AI ethics expertise. There is a great temptation to jump into the field, and to exploit opportunities that bring recognition from the community.²⁴ When things move fast, the traditional world of scientific publishing feels exceedingly slow. As a journal editor myself, I see firsthand how researchers might struggle as their studies of brand-new technologies are stuck in review and revision processes that mean their results will be outdated by the time they are finally published. However, I would argue that any manuscript that would *actually* be outdated because, for example, a software solution has been updated, is not appropriate for publication anyway, as it should contain analyses of the solution *and* considerations related to what could be learned and developed based on the results of the research. These should be valid and useful regardless of how rapidly the technology develops. Manuscripts that do not have these more timeless elements are perhaps better suited for other forms of dissemination, such as Medium or Substack posts. Or *preprints*?

One key difference between preprints and these other forms of publishing, however, is that preprints are more likely to stay available, accessible, and usable by other researchers than alternative formats. Preprints get a persistent identifier such as a DOI, and are indexed in scientific search engines. Another user benefit is that preprints cannot be changed without providing a record of changes. All of this makes the preprint a more solid form of dissemination likely to be used and referenced by other researchers, potentially helping promote the broader review and extensive discussion that I've discussed here.

In the following, I propose a tentative set of guidelines for preprinting, intended as a basis for community discussion and debate.

Consider alternative formats. Before preprinting, consider how to best disseminate your ideas to the relevant stakeholders. A key determinant of whether to preprint concerns the scholarly nature and ambition of the work. Only work that aspires to be scholarly or scientific research should be preprinted; this helps avoid the further di-

lution of preprinting as a valuable scientific practice. Much solid work is better suited for publication in mainstream or technology-specific magazines, blogs, or other outlets. Considering these alternative formats is important to reduce one's contribution to information overload in scientific channels.

Consider the need for speed. Are your ideas of a kind that will be immediately useful, and potentially even important, for stakeholders in the scientific community? Or is it important for you to establish precedence with your research—potentially without the need for broad dissemination? If the answer to either of these questions is yes, and if your work adheres to the relevant expected scientific standards, preprinting could be a good idea. If not, consider the downside of *too* rapid publication. The consequences of premature dissemination of research that does not *need* rapid publication could be that a) the ideas lose novelty, b) the ideas are discredited due to the immature state of the research, and, consequently, c) your reputation suffers.

Consider the need for broad review. If your ideas require a particularly extensive process of review and discussion, preprinting might allow for a broad “review” process before you finalize the research. However, this can also be achieved by using blogs or other outlets, as demonstrated by John Danaher, who uses his blog to publish partial and tentative analyses of his work.^c Some might also use mainstream media for testing and getting feedback on new ideas. This naturally requires that the research have a sufficiently broad appeal, or that researchers have access to journalists or others interested in aiding such efforts. If you preprint for broad review, make sure to state this clearly in the preprint, and spread it with an explicit request for feedback. Otherwise, one of the key benefits of preprinting is lost. As preprints are now used for so many different purposes, most of your peers will not realize that, just by posting a paper on a preprint server, you are looking for comments and feedback.

Consider the potential positive and negative consequences. How can

and will the preprint be used? If you see yourself as a free and isolated scientist with no responsibility for how others interpret and use your findings, this question might be irrelevant. Otherwise, you should know that your research will always have social implications, so be sure to consider the potential for your non-reviewed research to misdirect policy or practice. In the fields of AI and AI ethics, it will be particularly relevant to consider how your work might either contribute to AI hype *or* lead some to not adopt technologies that are societally beneficial. Such effects might be desired, but they could have major consequences and should ideally be the result of solid research that is not prematurely published.

Adhere to academic standards of research ethics when possible. AI is a field in which academics are by no means the only participants, or even the major ones. While researchers in universities and, for example, the healthcare sector tend to be subject to relatively strict standards of research ethics, this does not apply to AI ethicists working in tech companies. Researchers from companies such as Meta, Microsoft, and OpenAI can do things other researchers cannot, and though much industry research would be stopped at the gates by journals requiring adherence to ethical standards, these organizations are free to preprint seemingly scholarly work without any scholarly standards of ethics. This is a good argument for academic researchers to seek journal publication; however, it could also be a reason for industry researchers to voluntarily adhere to standards of ethics, to help their research gain broader acceptance and legitimacy. I argue that there are reasons for the establishment of research ethics standards and that doing so might be good regardless, and perhaps should even be mandated more broadly.

If you preprint: Engage, revise, and redact when appropriate. Many of the benefits of preprinting require you to actively disseminate your research and engage with those who read and comment on it. Ideally, this can and should lead to revisions of the work. And in instances where your audience discovers grave errors in your preprint,

c <https://philosophicaldisquisitions.blogspot.com>

it should be updated and, if possible, redacted. Without such engagement, preprinting is just a self-archival and self-promotion tool—most of the benefits are lost.

Consider your own role and authority. For established researchers who no longer feel they need to prove themselves through cumbersome peer review processes, it can be tempting to post *everything* as preprints. However, even the best among us might need some correction, and even if the publish-everything strategy might be successful in gaining attention and citations, raising oneself above one's community risks both hurting the community and, over time, undermining oneself. A related consideration is the disproportionate power wielded by authorities in the field to both raise up and hurt others. Considering impacts on others is important, especially for early-career researchers just starting out. Preprinting willy-nilly exacerbates the risks of arbitrarily using social community power.

Conclusion

A diverse field such as AI ethics is bound to have a number of fault lines separating various groups with different backgrounds. This, though, does not necessarily mean that the various groups are in conflict, or that they cannot learn from each other and cooperate.¹⁰ Bridging the fault lines, rather than shouting across the chasm without any real attempt to understand or constructively learn from and engage with each other, should be preferable. Fostering such cooperation, however, entails actively engaging in and developing community guidelines, coordination, and a joint understanding of these different backgrounds and perspectives.³⁸ This discussion of preprinting practices is an attempt to chip away at one small part of this challenge.

Preprinting can be highly beneficial—especially in AI ethics—but would benefit from adherence to certain guidelines to avoid some of the pitfalls. Even when these—or other, better—guidelines are followed, additional questions remain.

One important issue is how we can help each other navigate and leverage a rising tide of research of increas-

ingly unknown merit. There is no universally accepted definition of what constitutes scientific knowledge,²² but it is commonly identified through its recognition and appropriation by the scientific community.^{8,11} Developing venues for community debate and deliberation will be important to effectively make use of and curate preprints. It might also make sense to find, shape, and develop more interdisciplinary preprint servers for AI-ethics-related work, as existing servers are poorly suited for this purpose. Furthermore, these debates must include other stakeholders, including the media and citizens. Navigating research in the field is difficult enough for those in it, so making sure we help others understand what research can be trusted—for what purposes and how much—will be crucial as preprinting practices develop. 

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